

RaceDay

Documentation

Project documentation based on the current RaceDay project design

1. Application Overview

RaceDay is a full-stack, web-based event management system designed for the South African road running, walking and cycling community. The application provides a central platform for organisers and participants to manage sporting events, event categories, enrolments and race results. Its purpose is to reduce reliance on paper-based registration, spreadsheets and disconnected communication channels.

The system is designed around two main user groups: Organisers and Participants. Organisers manage the events they own, create event categories, view enrolments and capture results. Participants create accounts, browse upcoming events, enter an event by selecting a category, view their enrolments and access their personal result history.

The project is being developed progressively. The current design establishes the database, Entity Relationship Diagram (ERD) and REST API contract, with later implementation planned as a C# REST API followed by an MVC web application.

2. Purpose and Objectives

- Provide a central online system for managing sporting events.
- Allow organisers to create, update and delete events that they own.
- Allow organisers to create and manage event categories.
- Allow participants to discover upcoming events and enrol in categories.
- Maintain structured participant enrolment records.
- Allow organisers to capture and correct participant race results.
- Allow participants to view their historical results.
- Provide public event information and event leaderboards where permitted.
- Use role-based access control for organiser and participant operations.
- Maintain a relational database supporting the application's requirements.

3. Target Users

User Type	Main Responsibilities	Key Features
Organiser	Create/manage owned events; manage categories; view entrants; record results.	Event management, categories, enrolments, results
Participant	Register, browse events, enter categories and track race history.	Registration/login, browsing, enrolment, personal results
Public Visitor	View event information and public results.	Browse events, event details/categories, leaderboards

4. Core Features

4.1 User Registration and Authentication

RaceDay provides account registration and login. Accounts can be created as Organisers or Participants. Participant registration can include a date of birth. Successful login issues a JWT access token for protected operations.

- Register an Organiser or Participant.
- Validate required registration information.
- Prevent duplicate email registration.
- Log in using email and password.

- Issue a JWT access token after successful authentication.
- Reject missing or invalid authentication tokens.

4.2 User Profile Management

Logged-in users can retrieve and update their own profile information.

- View the current user's profile.
- Update profile information such as full name.
- Allow participant-specific date-of-birth updates.
- Protect profile operations using authentication.

4.3 Event Management

Events are the central objects in RaceDay. Each event belongs to an organiser and stores its name, description, date, location, distance and event type.

- Browse upcoming events.
- Filter events by date range or location.
- View an event with its categories.
- Organisers create events.
- Owning organisers update events.
- Owning organisers delete events and their categories.
- Ensure organisers can modify only events they own.

4.4 Event Categories

Categories provide different participation options within an event, such as a Half Marathon or 10km Fun Run.

- Publicly view event categories.
- Owning organisers add categories.
- Owning organisers edit category names.
- Owning organisers delete permitted categories.
- Prevent deletion when a category already has enrolments.

4.5 Participant Enrolment

Participants enter an event by selecting a category. The authenticated participant is taken from the login token, and duplicate enrolments in the same category are prevented.

- Enter through a selected category.
- Automatically associate the authenticated participant.
- Store event and category associations.
- Prevent duplicate category enrolments.
- View personal enrolments.
- Withdraw from an owned enrolment.
- Allow the owning organiser to view event enrolments.

4.6 Race Results

Organisers can record a participant's finish time and optional finishing position. Existing results can be corrected or updated.

- Capture finish time.
- Record optional finishing position.
- Prevent duplicate results for one enrolment.
- Update an existing result.
- View personal result history.
- Provide a public event leaderboard with participant, category, finish time and position.

5. Typical Application Workflow

1. An organiser creates an account and logs in.
2. The organiser creates a sporting event.
3. The organiser adds categories.
4. A participant creates an account and logs in.
5. The participant browses events and views categories.
6. The participant selects a category and enrolls.
7. The organiser views event enrolments.
8. After the event, the organiser records finish time and position.
9. Participants view their results and the public event leaderboard can display results.

6. System Data Model

The current ERD contains six entities: Organisers, User, Event, Category, Enrolment and Result. Primary keys identify records and foreign keys connect related records.

Entity	Purpose	Important Data
Organisers	Stores organiser accounts that own events.	OrganiserID, FullName, Email, PasswordHash
User	Stores participant-facing accounts.	UserId, FullName, Email, PasswordHash, Role, DateCreated
Event	Stores sporting events.	EventID, OrganiserID, EventName, Description, EventDate, Location, Distance, EventType
Category	Stores event participation categories.	CategoryID, EventID, CategoryName
Enrolment	Links participants to event categories.	EnrolmentID, UserId, EventID, CategoryId, EnrolmentDate
Result	Stores completed enrolment results.	ResultID, EnrolmentID, FinishTime, Position

The current database design intentionally keeps both EventID and CategoryID in Enrolments, matching the reviewed project ERD.

7. Relationship Summary

Relationship	Meaning
Organiser → Event (1:*)	One organiser can create and own multiple events.
Event → Category (1:*)	One event can offer multiple categories.
Event → Enrolment (1:*)	An event can have multiple participant enrolments.
Category → Enrolment (1:*)	A category can contain multiple enrolments.

Relationship	Meaning
User → Enrolment (1:*)	One participant can have multiple enrolments.
Enrolment → Result (1:1)	An enrolment can have one recorded result.

8. REST API Features

The planned REST API is divided into Authentication, User Profile, Events, Categories, Event Enrolments and Results. The API plan defines methods, routes, descriptions, roles, request bodies and expected responses.

Area	Example Operations	Access
Authentication	Register; login	Public
User Profile	View/update current profile	Logged-in user
Events	List; view; create; update; delete	Public viewing; Organiser management
Categories	List; add; update; delete	Public viewing; owning Organiser management
Enrolments	Enter; list own; view event entries; withdraw	Participant/owning Organiser
Results	Create; update; history; leaderboard	Depends on operation

9. Security and Access Control

RaceDay uses role-based access control. Public endpoints do not require login, while protected endpoints require an authenticated user. Organiser-only operations are restricted to authenticated organisers, and owner checks ensure an organiser owns the relevant event or category before modifying it.

- JWT-based authentication is planned.
- Passwords are represented as password hashes rather than plain text.
- Users access their own profile information.
- Organisers manage only resources they own.
- Participants withdraw only their own enrolments.
- Invalid authentication can return HTTP 401.
- Unauthorised ownership attempts can return HTTP 403.
- Duplicate enrolments/results can return conflict responses.

10. Development Roadmap

Part	Planned Work
Part 1 – Planning	ERD, REST API endpoint plan and SQL Server database script.
Part 2 – API	Build the C# RESTful API, connect it to the database, add unit tests and GitHub Actions CI/CD.
Part 3 – Web Application	Build the MVC web application, integrate Azure Blob Storage and containerise with Docker.

11. Current Scope and Limitations

This documentation describes the functionality supported by the supplied RaceDay project design. The current specification does not define online payments, automated timing hardware, live checkpoint tracking, messaging,

sponsorship management or race-pack logistics. These should not be treated as current features unless added to the project specification.

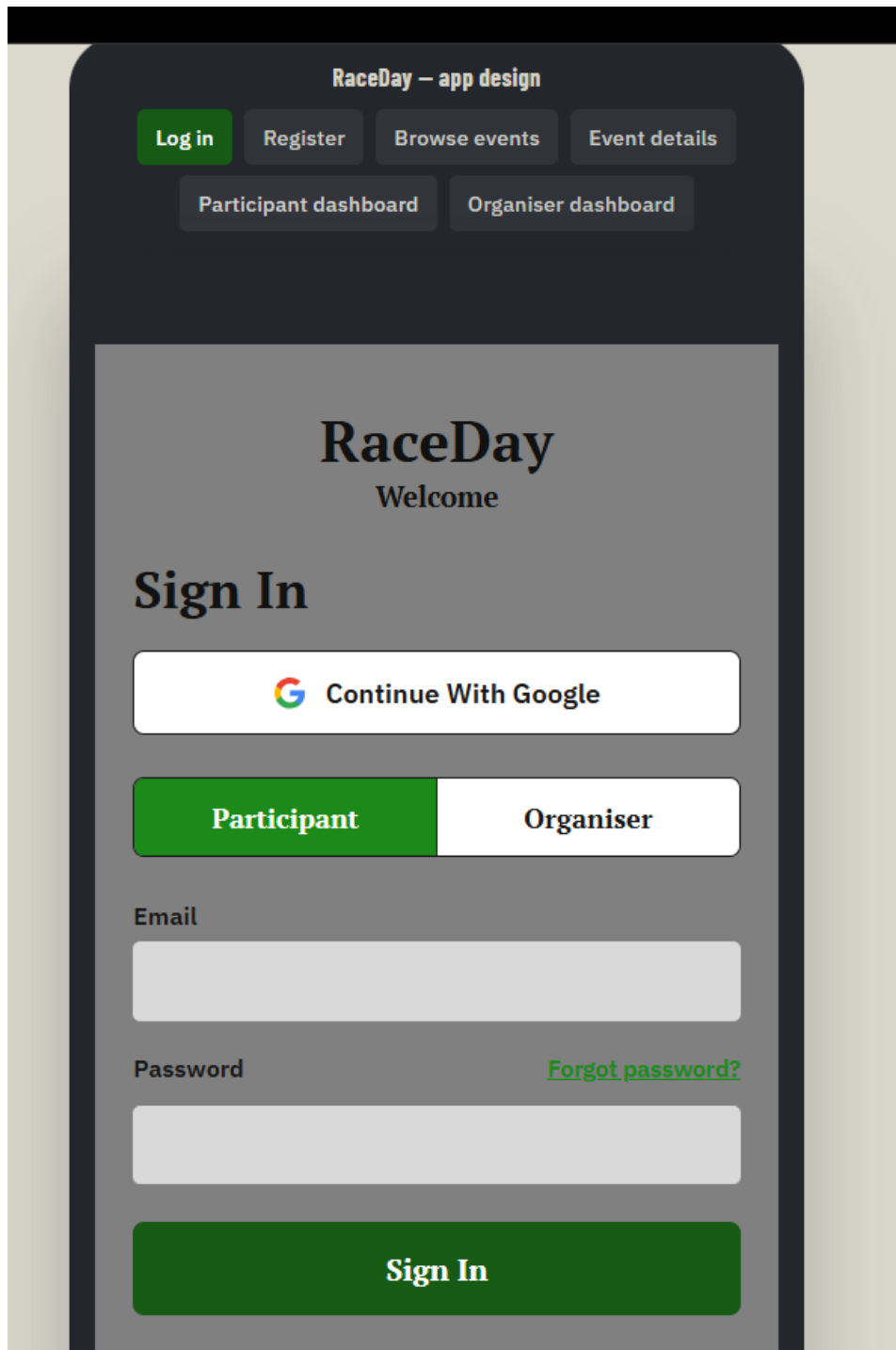
12. Conclusion

RaceDay provides a structured foundation for managing sporting events from event creation through participant enrolment and results. The separation between organisers and participants, combined with role-based API access and a relational SQL Server database, provides a clear architecture for the planned C# REST API and MVC web application.

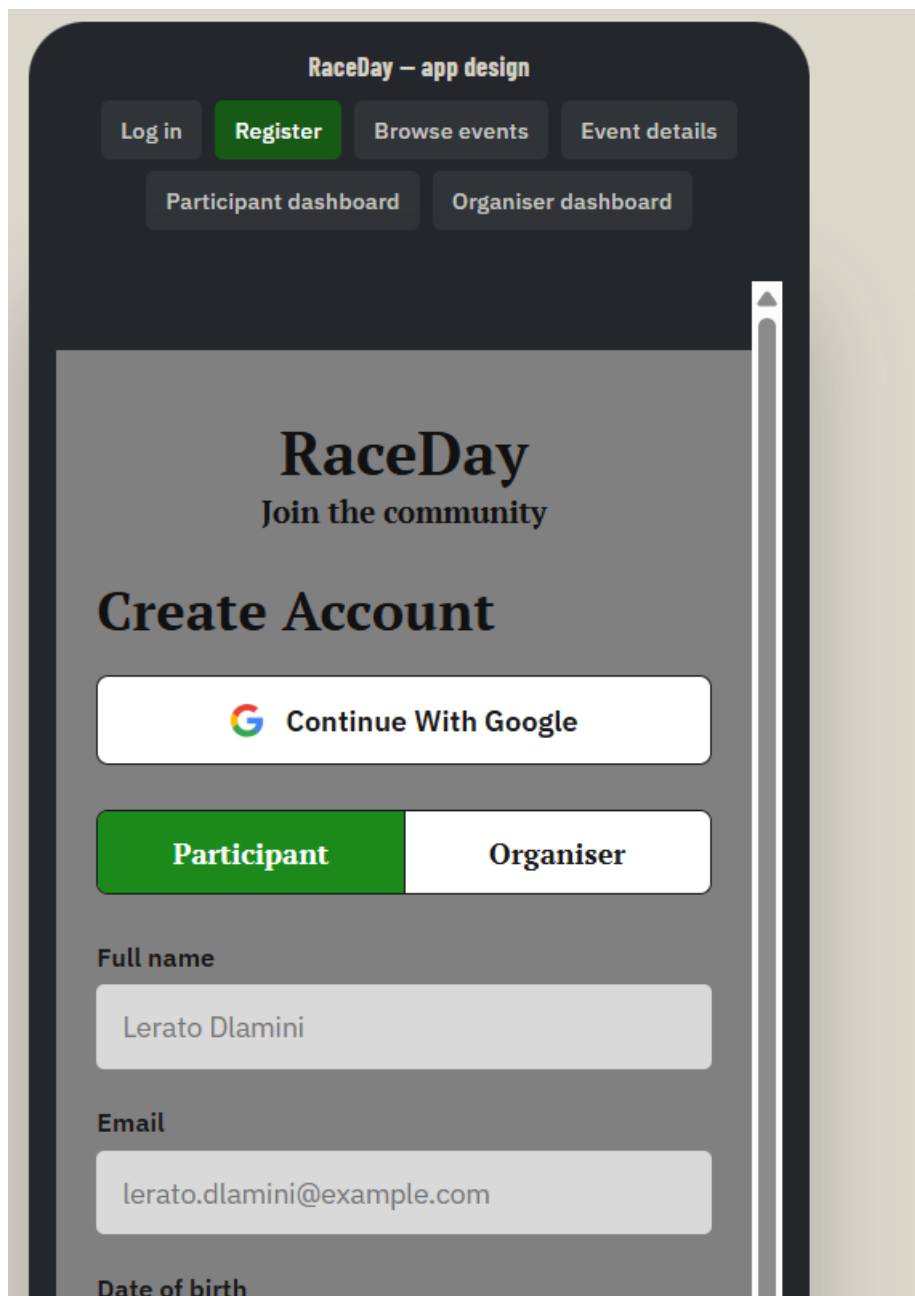
13. Project Documentation Sources

- RaceDay README – project overview, roles, project phases, ERD/database description and setup information.
- RaceDay API Endpoint Plan – authentication, profile, event, category, enrolment and result functionality.
- RaceDay ERD (UML notation) – entities, attributes, keys and relationships.
- RaceDay SQL Server database schema – table definitions, constraints and seed data

Design preview



This is the login screen for registered users



Create Account



Continue With Google

Participant

Organiser

Full name

Lerato Dlamini

Email

lerato.dlamini@example.com

Date of birth

yyyy/mm/dd



Required for participants, used for age-category results.

Password

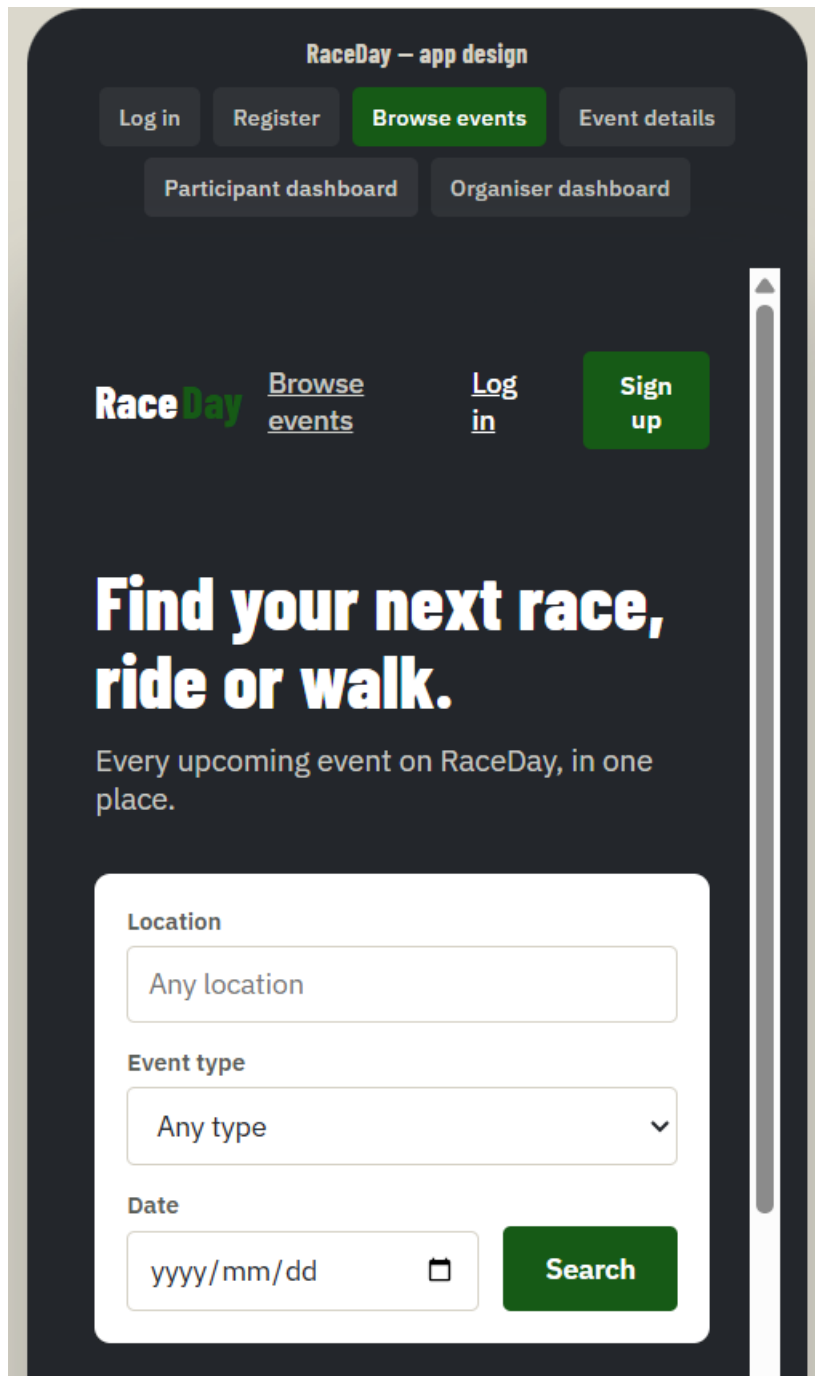
At least 8 characters

Create Account

Already have an account? [Log in.](#)

By creating an account you agree to our [Terms of Use](#) and
POPIA Privacy Policy

Those who haven't registered



3 upcoming events

Walking / running

20 Sep 2026

Durban Beachfront Park Run Challenge

Durban, KwaZulu-Natal



5km & 10km categories

[View event](#)

Cycling

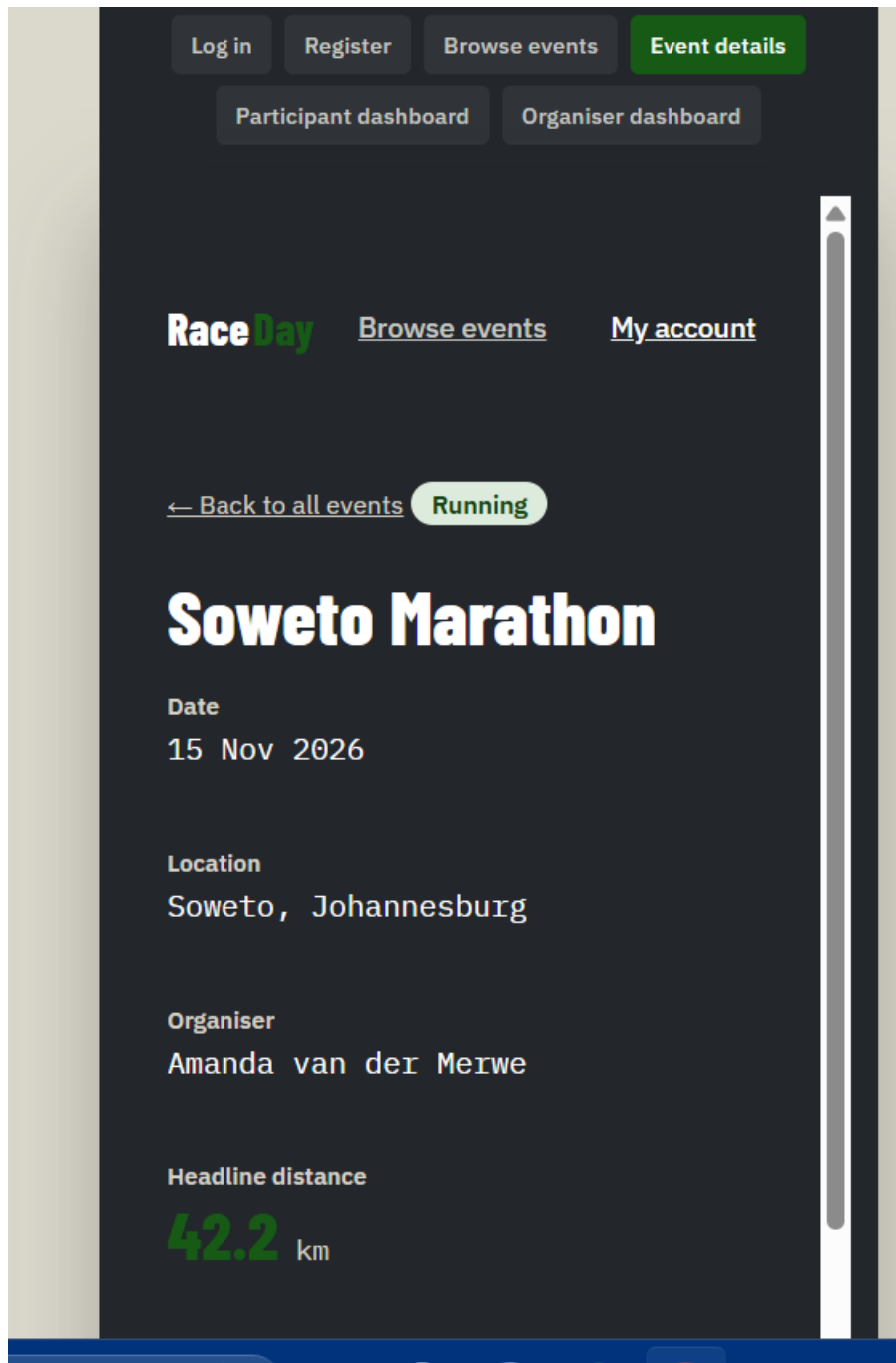
8 Nov 2026

Cape Town Cycle Tour Community Ride

Cape Town, Western Cape



Looking for an event to register



About this event

An iconic road running event through the historic streets of Soweto. Choose your category below — entries close two weeks before race day.

Categories

Full Marathon

42.2 km

Enter

Half Marathon

21.1 km

Enter

10km Fun Run

10.0 km

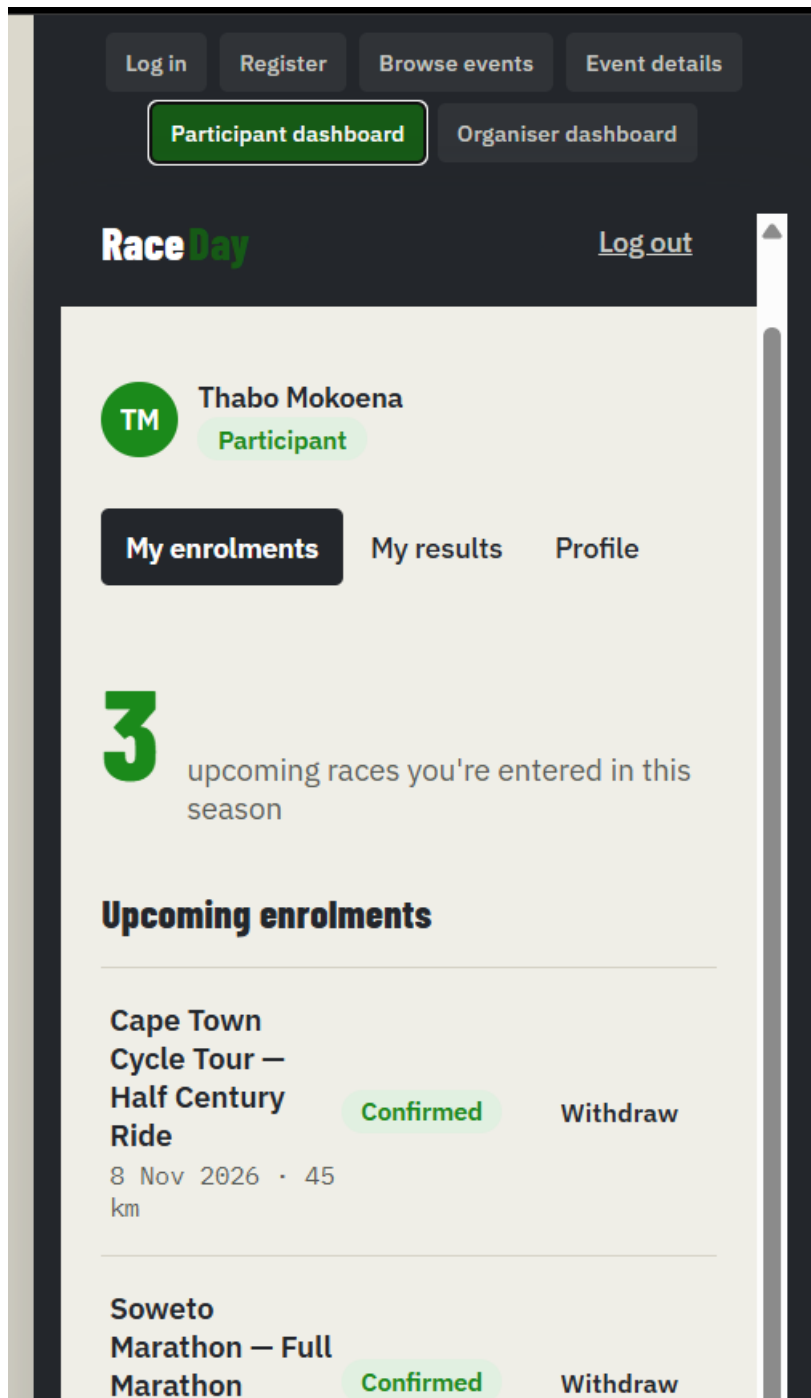
Enter

Route

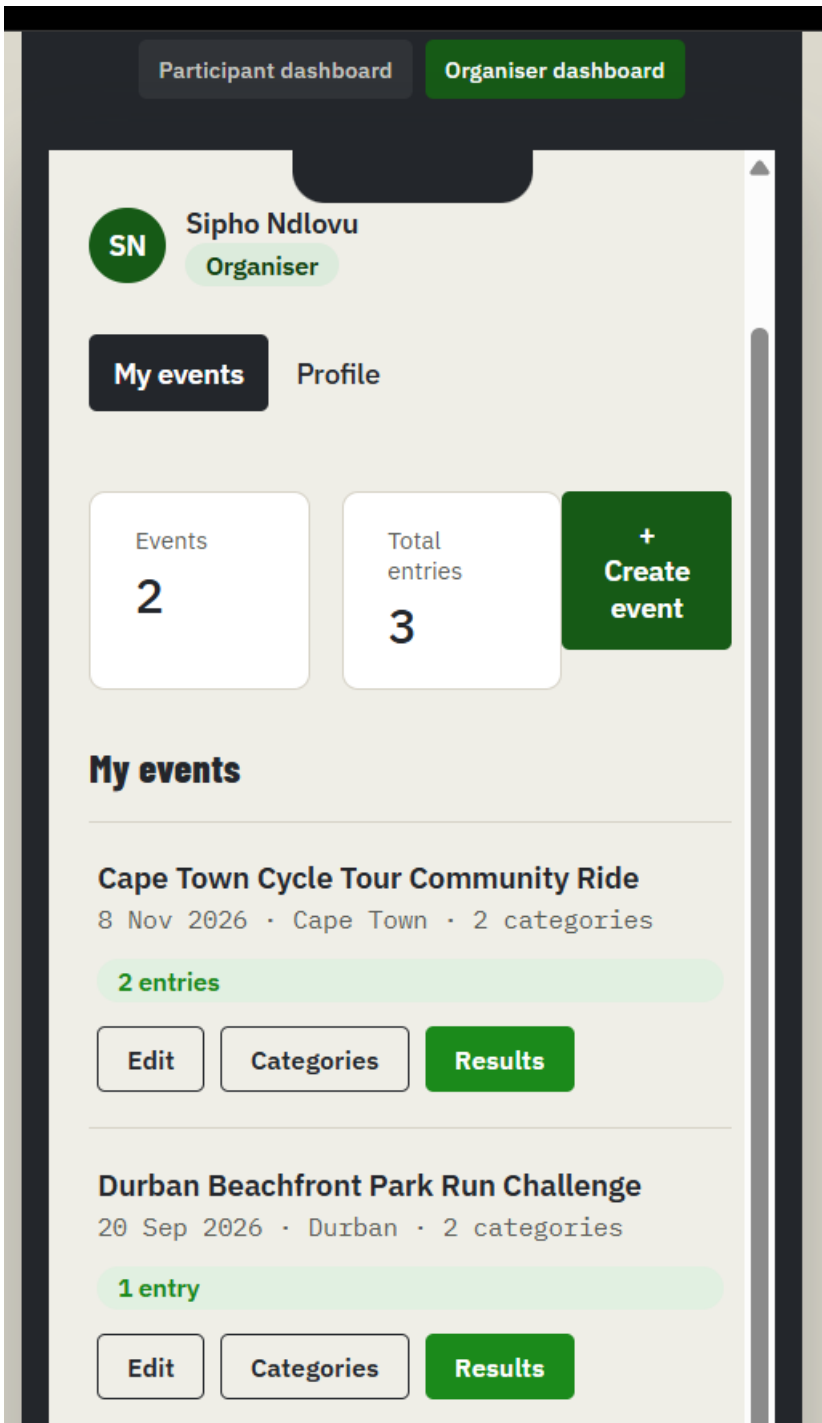


Route preview — served from the Map API once

Viewing the event details



Details of the participants/ user's enrolments



Capture a result – Durban Beachfront, 10km Challenge

Participant

Thabo Mokoena – bib 3002



Finish time

00:48:12

Position

7

Save

The organizer/ admin can edit events, capture results and create an event